Models with a Single Categorical Predictor (One-Way ANOVA).

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

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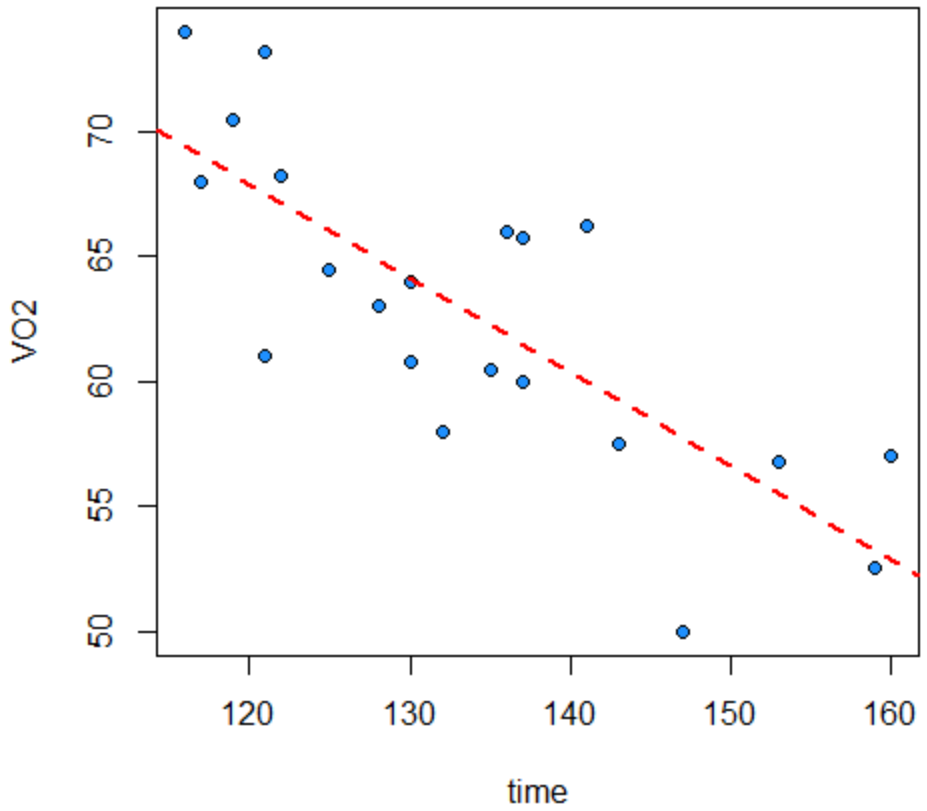
Relationship between $VO2_{max}$ and 800 meter times.



$$\widehat{VO2_i} = 112.89 - 0.37(TIME_i)$$

The intercept was statistically significant, $t(19) = 12.19$, $p < 0.05$, and the slope was statistically significant, $t(19) = -5.44$, $p < 0.05$.

800m run times explained considerable variance in $VO2_{max}$ (PRE = 0.61).

Relationship between $VO2_{max}$ and 800 meter times.

- But what if we had groups of runners (faster versus slower) rather than continuous run times?
  - Our **continuous** variable becomes discrete and **categorical**.

	sudID	VO2	time	faster
1	s001	74.00	116	1
2	s002	68.00	117	1
3	s003	70.50	119	1
4	s004	61.00	121	1
5	s005	73.25	121	1
6	s006	68.25	122	1
7	s007	64.50	125	1
8	s008	63.00	128	1
9	s009	60.75	130	1
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15	s015	65.75	137	0
16	s016	66.25	141	0
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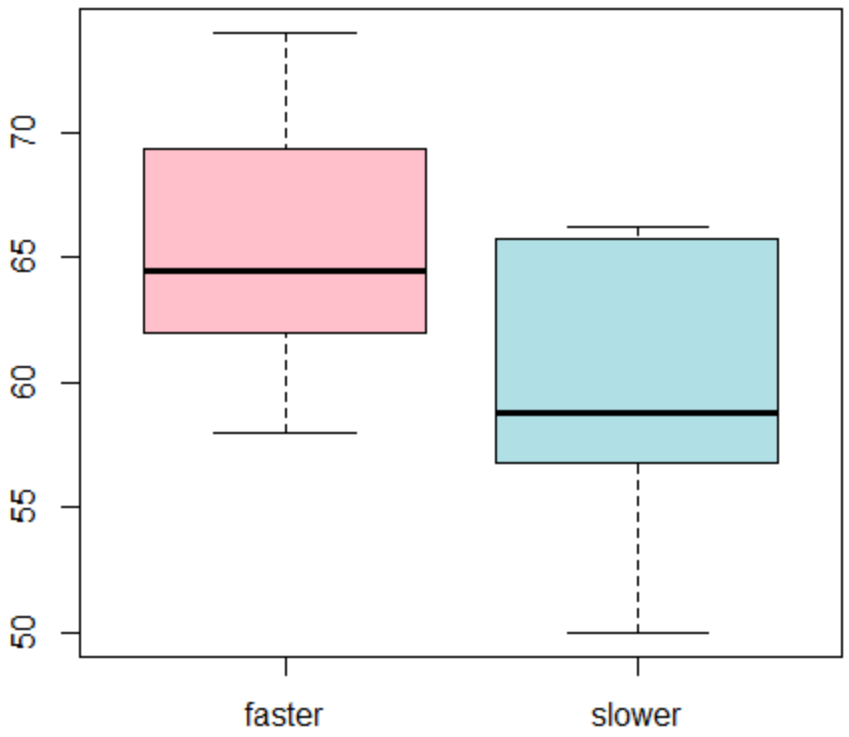
Categorical (2-level) variables in a GLM



- If ‘slower’ is coded as ‘0’ and faster is coded as ‘1’, what is our model?

$$\hat{y}_i = 59.225 + 6.707(faster_i)$$

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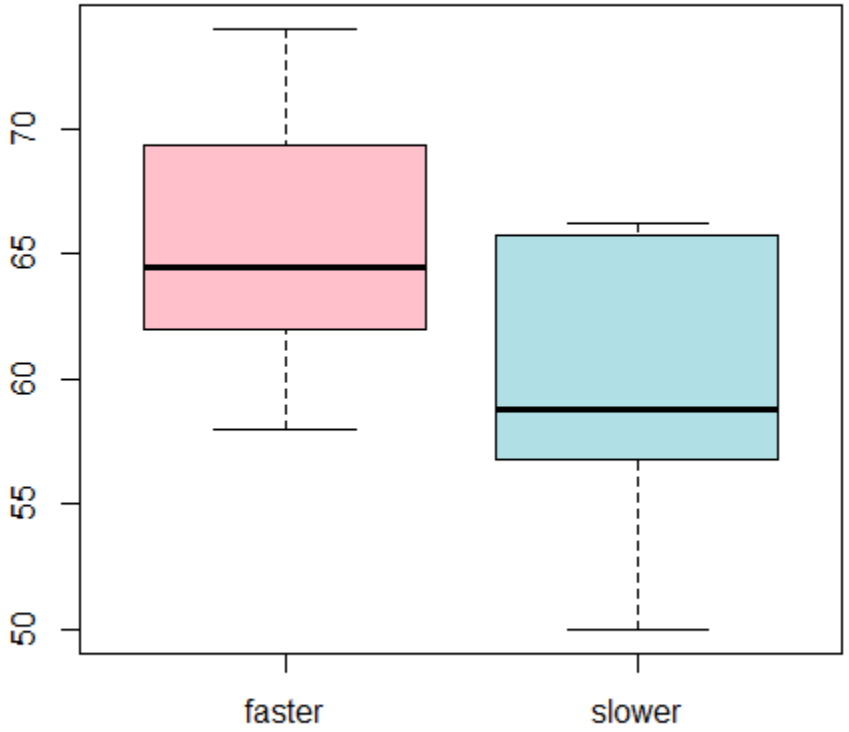


```
Call:
lm(formula = VO2 ~ faster, data = dat)

Residuals:
   Min     1Q Median     3Q    Max 
-9.225 -2.932 -1.432  4.568  8.068 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   59.225      1.720  34.434   <2e-16 ***
faster         6.707      2.376   2.822   0.0109 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.439 on 19 degrees of freedom
Multiple R-squared:  0.2954,	Adjusted R-squared:  0.2583 
F-statistic: 7.965 on 1 and 19 DF,  p-value: 0.01088
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What if we change how we code ‘slower’ and ‘faster’?

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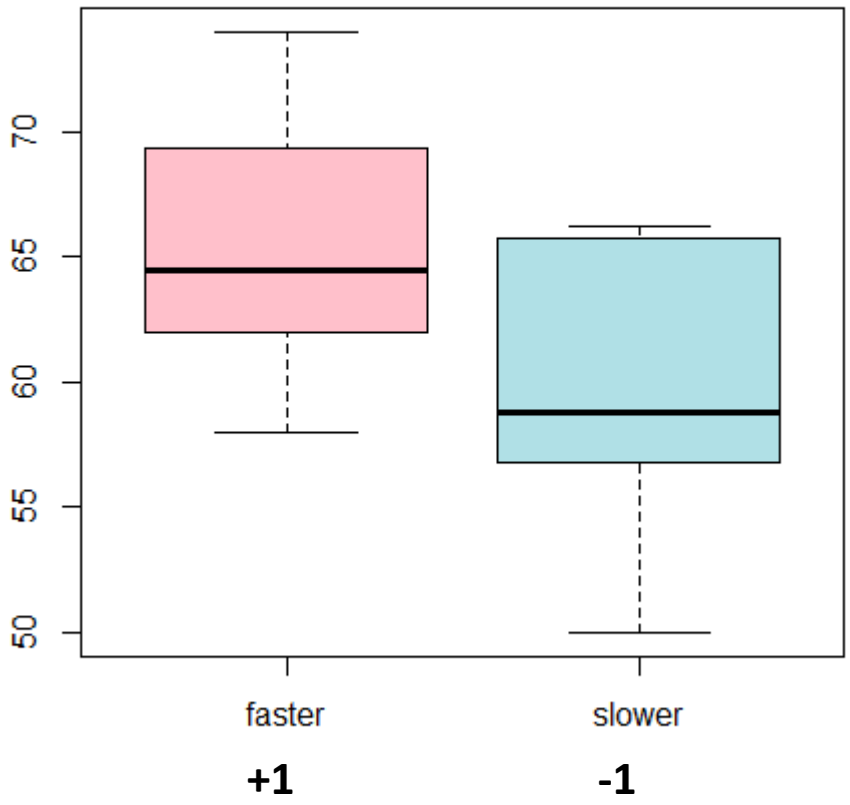

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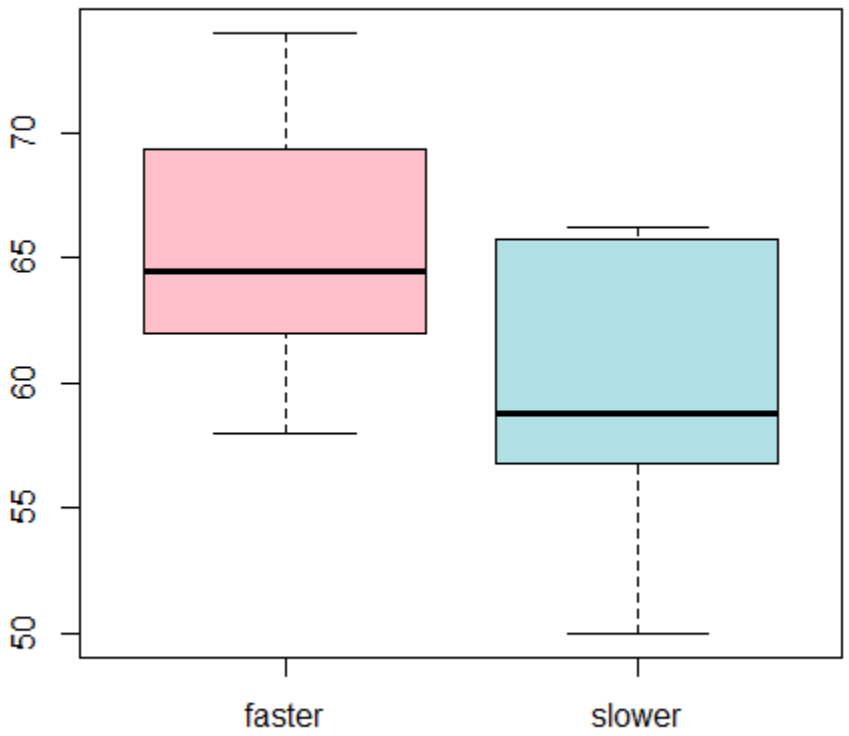
- What is 62.578?

- What is 3.353?



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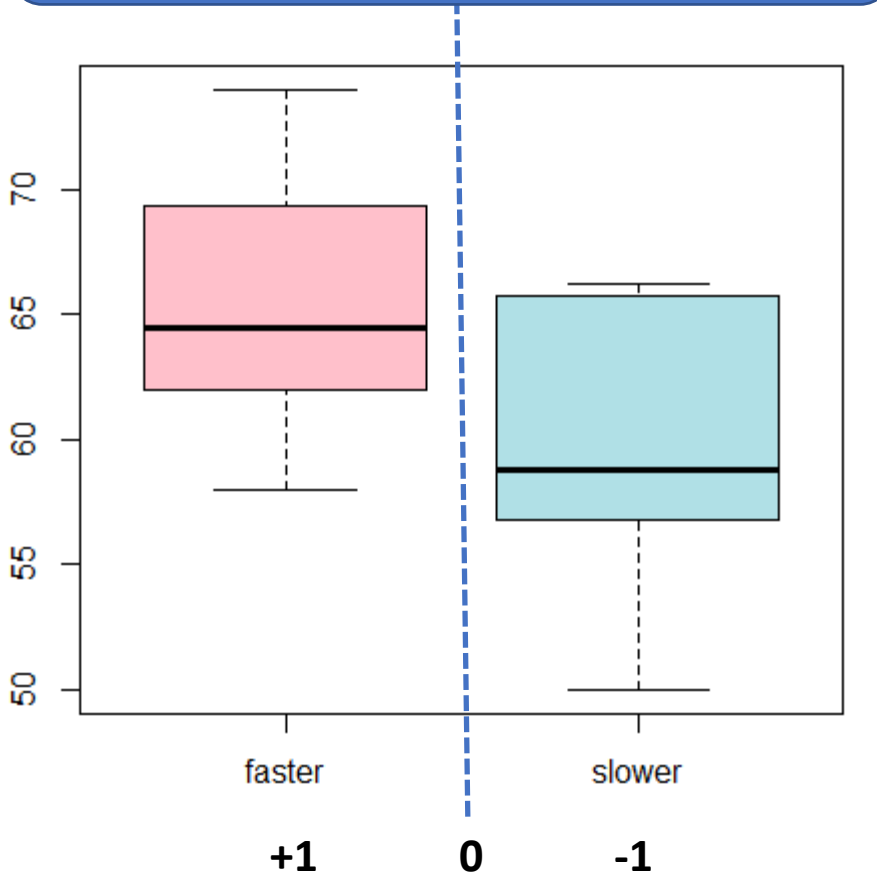

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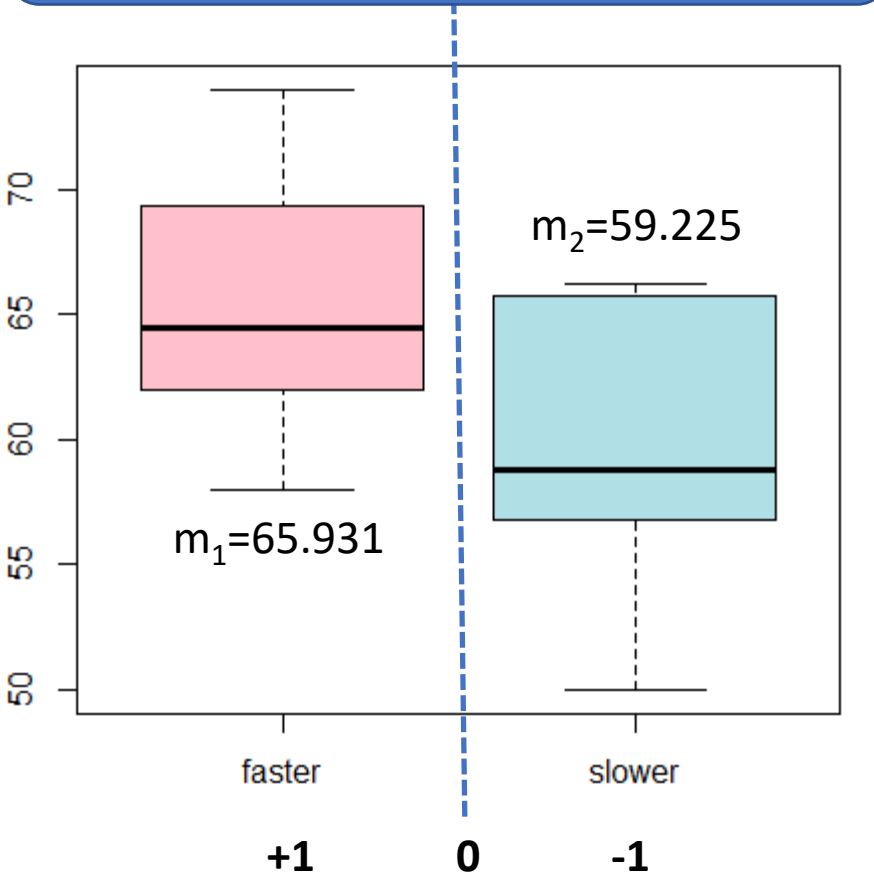
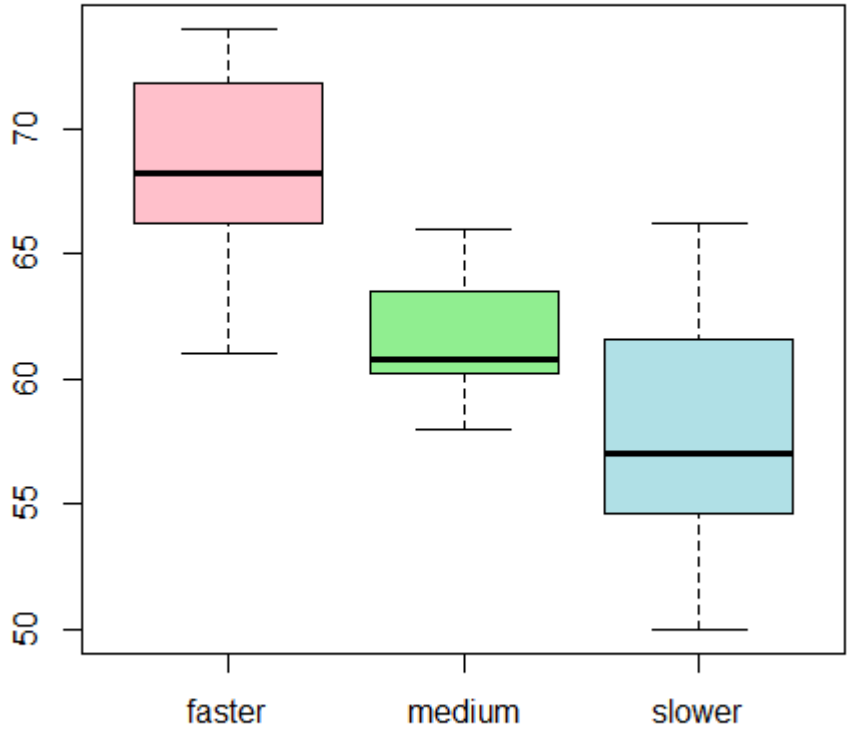


What if we have more than 2 groups?

Categorical (k > 2) variables in a GLM



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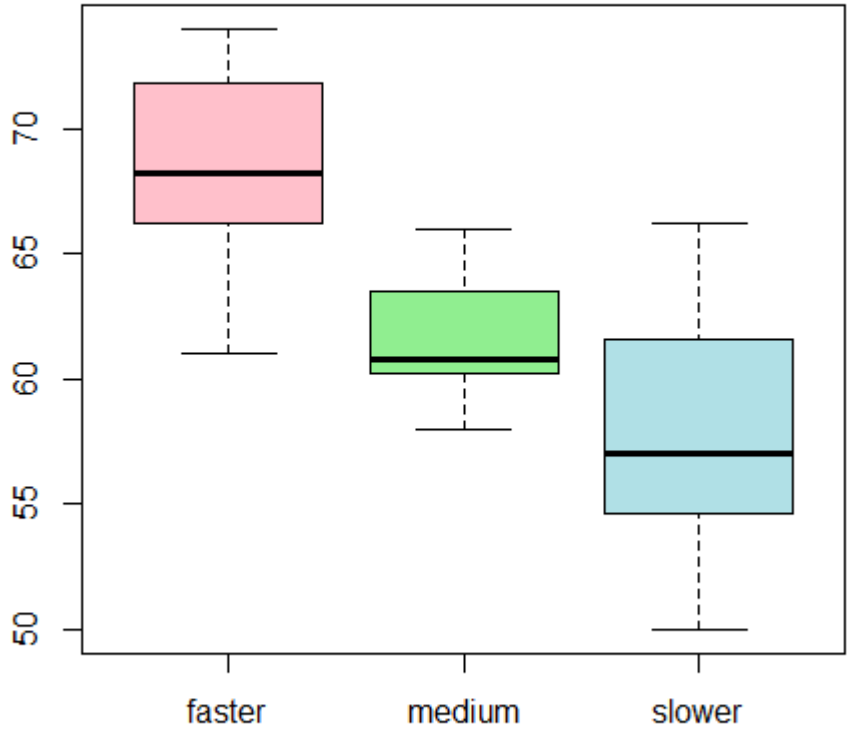

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- By default R uses 'dummy' codes:
  - λ1<-c(0,1,0)
  - λ2<-c(0,0,1)
  - What would a one unit change mean for these codes?

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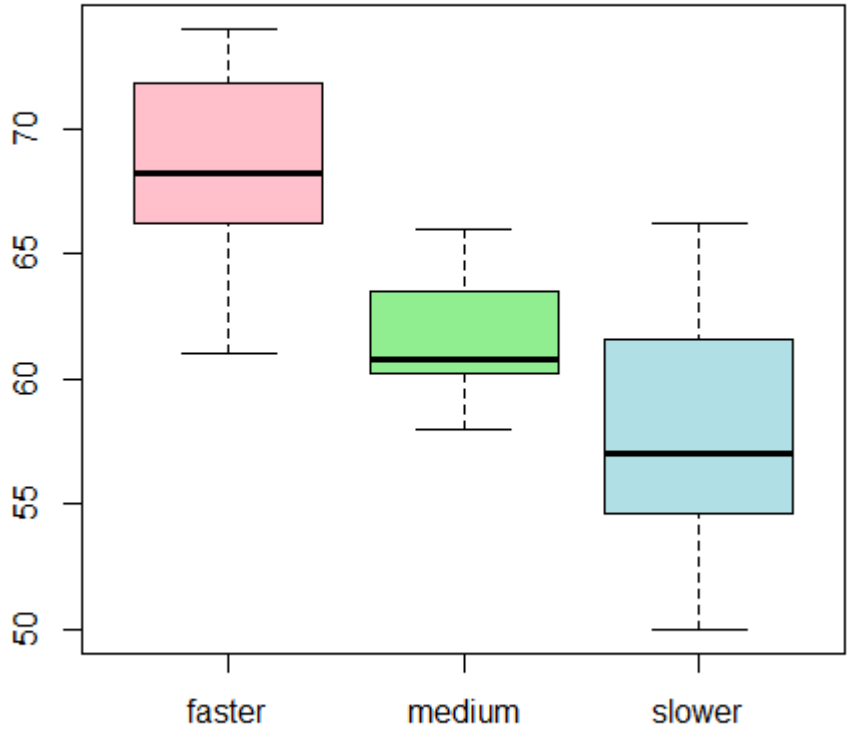


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Residuals:
   Min      1Q Median      3Q     Max
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Coefficients:
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(Intercept)    68.500      1.779  38.497  < 2e-16 ***
groupsmedium   -6.750      2.516  -2.682 0.015209 *
groupsslower  -10.536      2.516  -4.187 0.000554 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.708 on 18 degrees of freedom
Multiple R-squared:  0.4999,  Adjusted R-squared:  0.4443
F-statistic: 8.996 on 2 and 18 DF,  p-value: 0.001957
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Med	61.75	2.71
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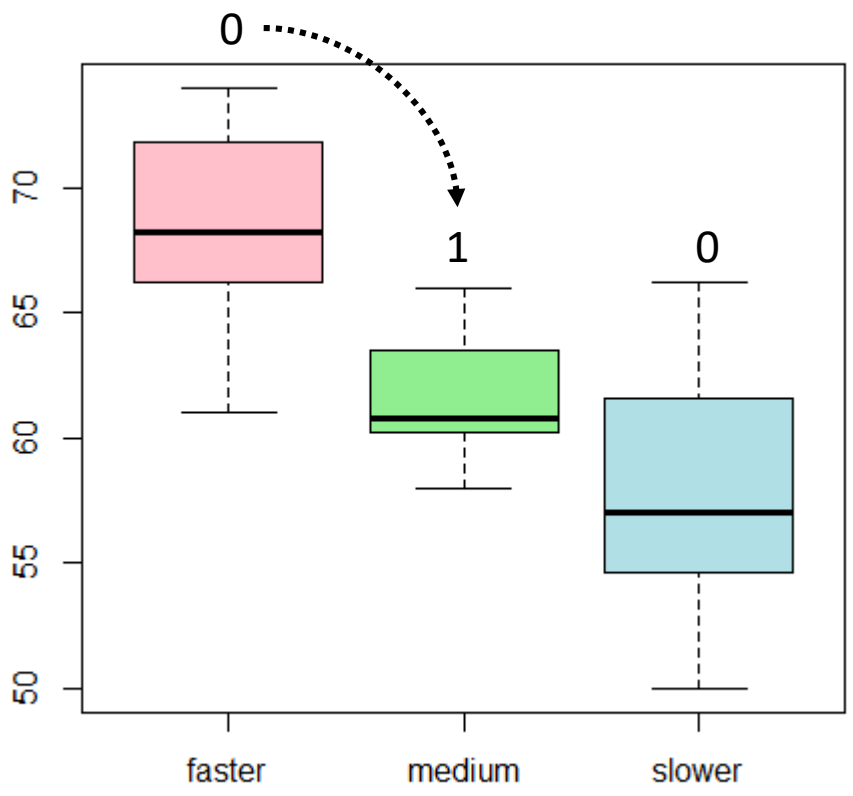


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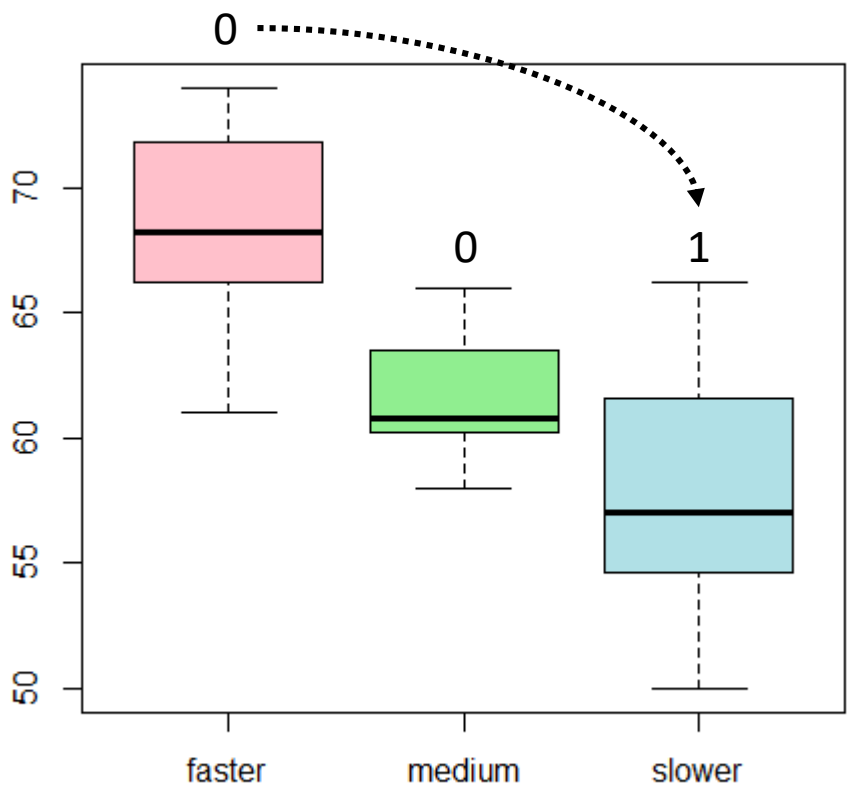


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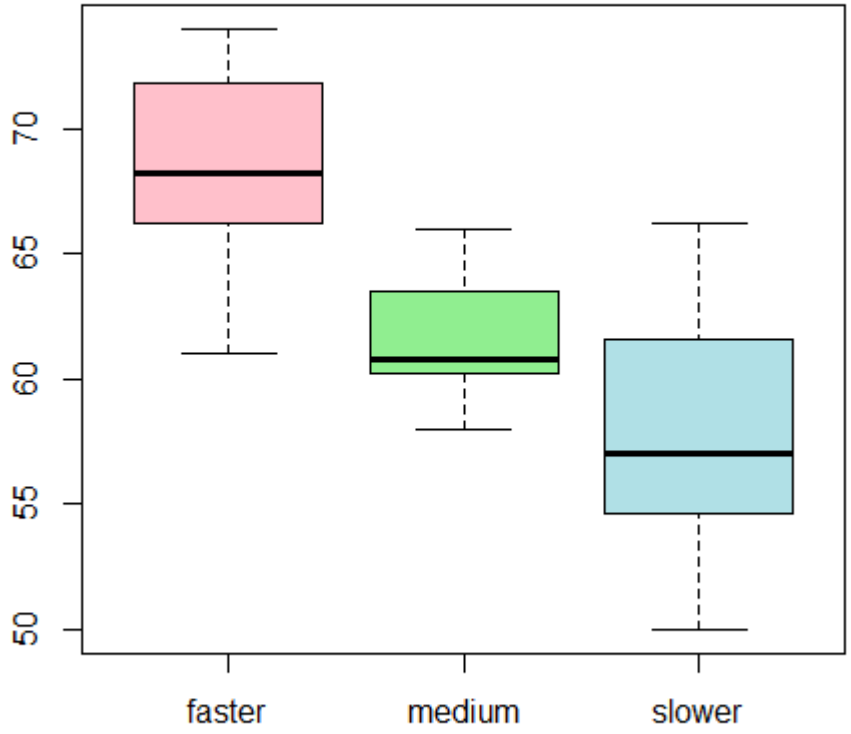

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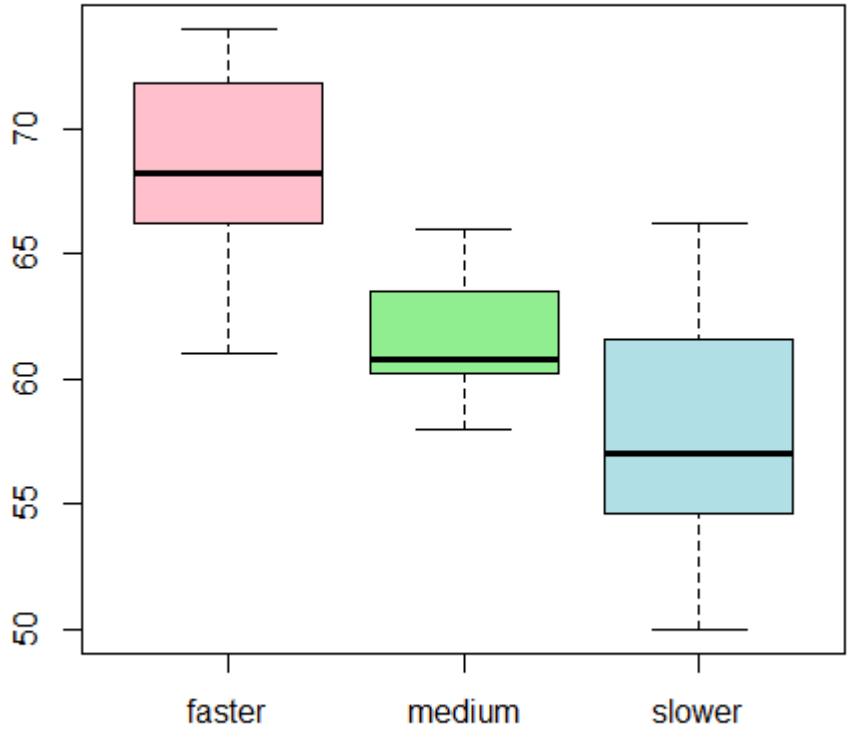

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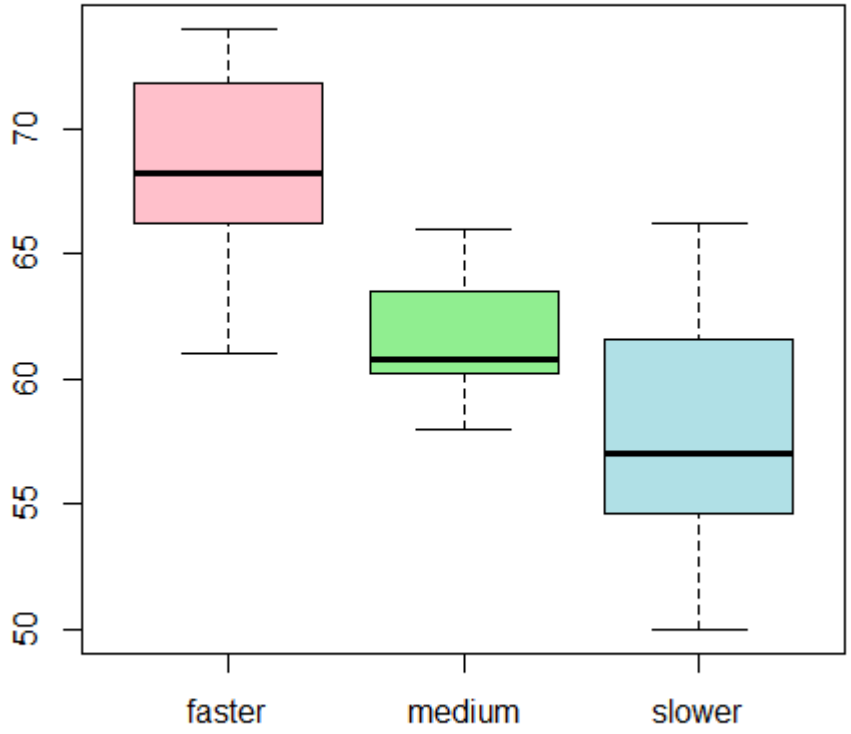


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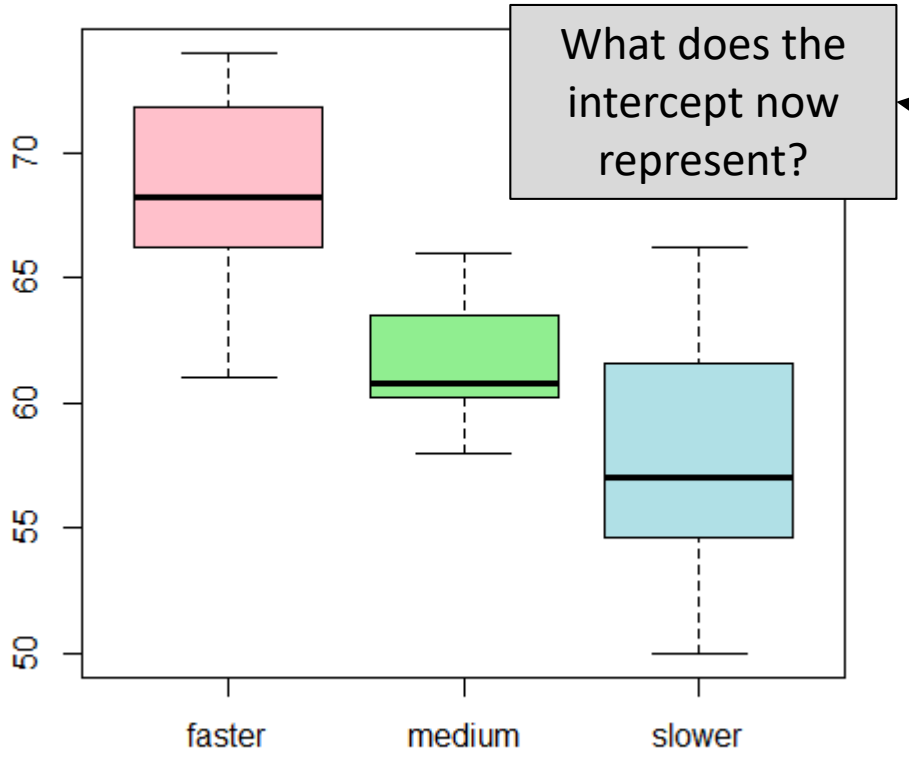


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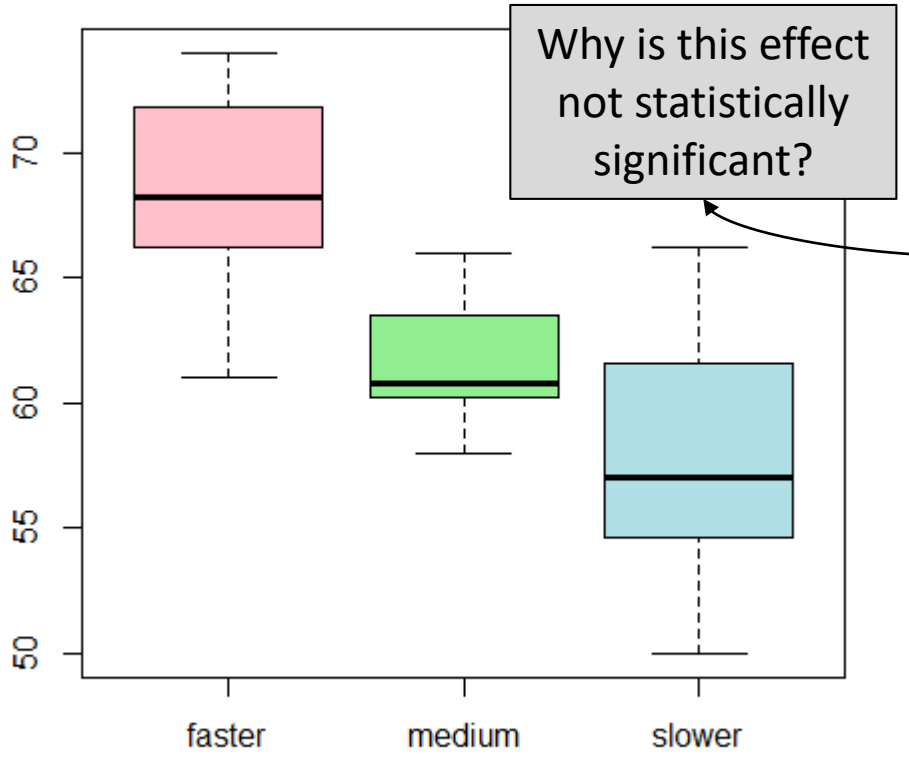


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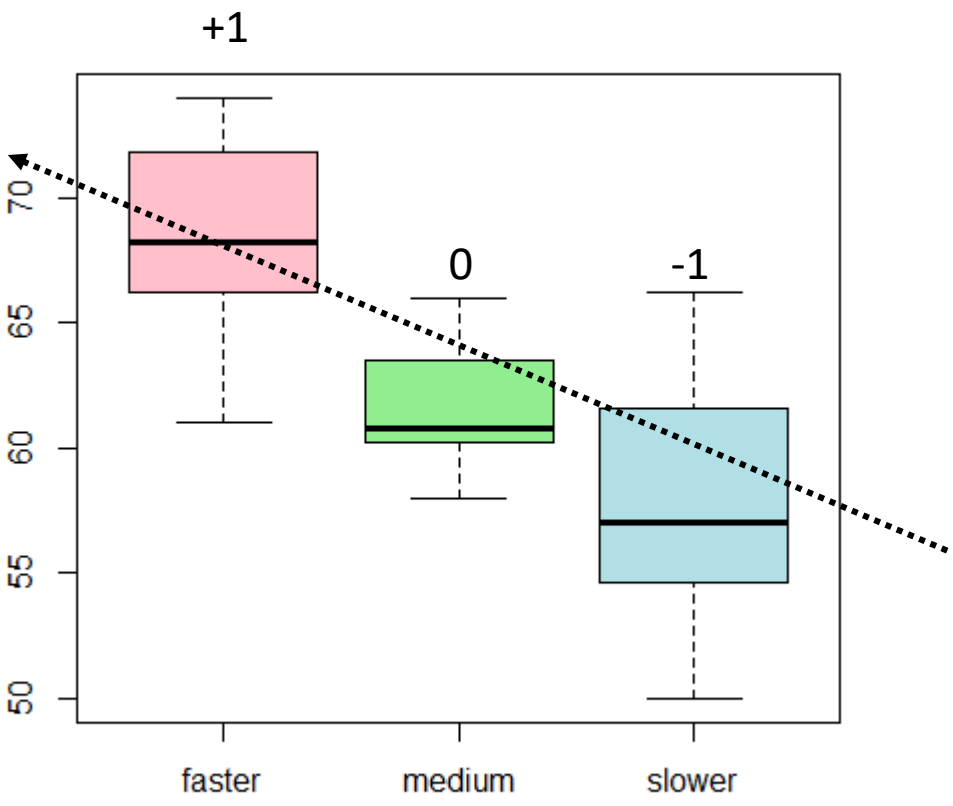


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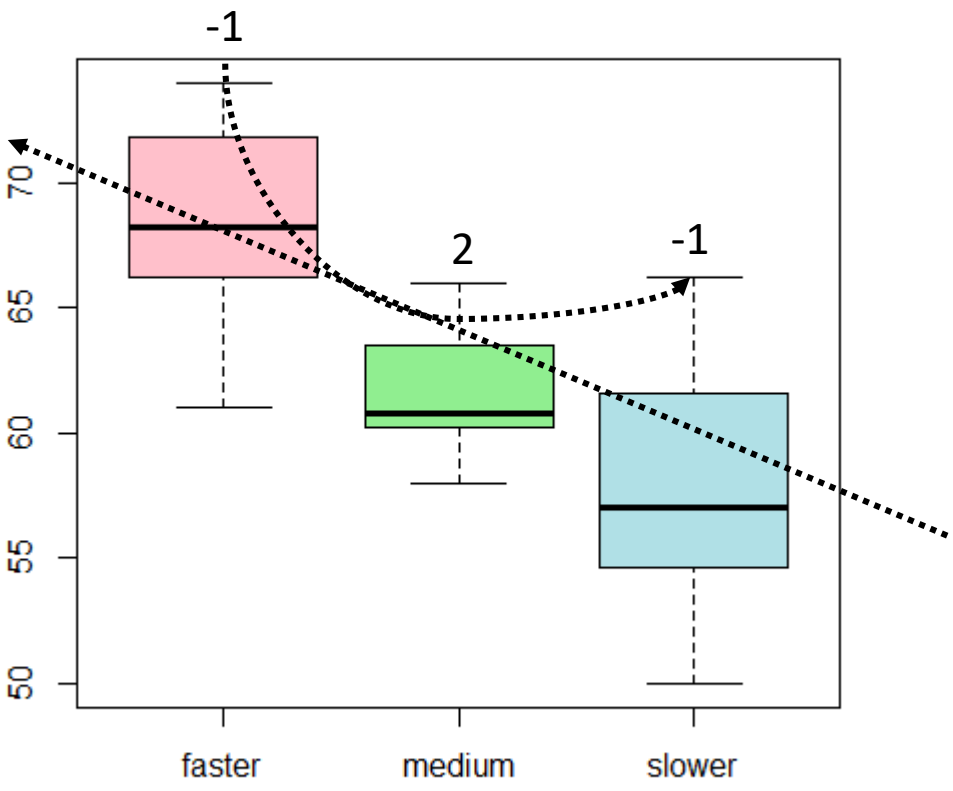


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F-statistic: 8.996 on 2 and 18 DF,  p-value: 0.001957
```

Group	Mean	SD
Fast	68.5	4.64
Med	61.75	2.71
Slow	57.96	6.13

Comparing our Different Codes

Treatment or 'Dummy' Codes

```
Residuals:
   Min     1Q Median     3Q    Max 
-7.964 -1.750 -0.500  2.250  8.286 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)    68.500      1.779  38.497  < 2e-16 ***
groupsmedium   -6.750      2.516  -2.682 0.015209 *  
groupsslower  -10.536      2.516  -4.187 0.000554 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.708 on 18 degrees of freedom
Multiple R-squared:  0.4999,	Adjusted R-squared:  0.4443 
F-statistic: 8.996 on 2 and 18 DF,  p-value: 0.001957
```

Contrast Codes

```
Residuals:
   Min     1Q Median     3Q    Max 
-7.964 -1.750 -0.500  2.250  8.286 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  62.7381     1.0273  61.070  < 2e-16 ***
lin.c         5.2679     1.2582   4.187 0.000554 ***
quad.c       -0.4940     0.7264  -0.680 0.505086    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.708 on 18 degrees of freedom
Multiple R-squared:  0.4999,	Adjusted R-squared:  0.4443 
F-statistic: 8.996 on 2 and 18 DF,  p-value: 0.001957
```